

CiteAs: Better Software through Sociotechnical Change for Better Software Citation

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Examples: <http://yt-project.org> <https://cran.r-project.org/web/packages/stringr> [More examples](#)

Figure 1: The landing page of CiteAs.org

ABSTRACT

Research increasingly depends on software, yet software work is often relatively invisible to the reputation economy shaping research careers. To address this, we demonstrate CiteAs, a system designed to improve incentives for high-quality software work by altering citation behaviors in the loosely governed sociotechnical system producing research software. CiteAs has two goals: first, to encourage those producing research software to make clear requests for citations most valuable to them; and second, to encourage users of research software to cite software more and follow the citations requested by the developer. CiteAs is implemented as a specialized search engine at <https://citeas.org/> that explains itself and instructs

better practices. Ultimately we hope that CiteAs will increase the visibility of research software, improving incentives for the software work needed to advance research.

CCS CONCEPTS

• **Human-centered computing** → **Collaborative and social computing systems and tools.**

KEYWORDS

citation, software production, incentives, persuasive technology

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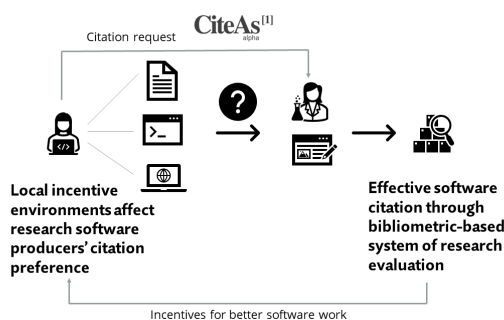


Figure 2: How CiteAs works

1 INTRODUCTION

It is hard to imagine conducting research without using research software [5]. However, the evaluation of research impact has conventionally been accomplished through bibliometrics based on formal citations; and research software has not been consistently cited in the research literature, dampening its impact. As a result, there exist weak incentives for researchers to take on the unpaid work of developing and maintaining research software [1, 10, 11], nor are there strong demonstrable arguments for institutions or funders to support research software. Research software can therefore be weakly supported and can struggle to realize its full potential to advance research [6–8]. If software contributions were more visible, the software work would be more likely to generate reputational rewards for its contributors, leading to increased time and money to build better software advancing research frontier.

Implementing software citation in the loosely governed scholarly ecosystem is a complex sociotechnical challenge. It calls for technical changes in standards, applications, and infrastructures, such as new metadata formats and indexing services for software citation; it also requires coordinated efforts in the uptake of emerging standards and integration of new practices by global stakeholders, including scholars, technologists, funders, research institutions and infrastructure maintainers. [9]. Prior interviews reveal that it could be paradoxical for individuals to act on software citation: developers of research software may want their “software papers” or research articles to be cited instead of the software itself, because it can help them make a better case during grant writing, academic job seeking, or applying for tenure [8]. Therefore, the challenge is sociotechnical: the desires of producers responding to their local incentive environment are just as relevant as technical specifications such as standardized identifiers and guidance for software citation (e.g., [9]). CiteAs, then, is a specialized search engine designed to prompt research software developers to actively create conditions for effective crediting by making clear citation requests, while channeling their requests to other researchers when they are preparing publications (Figure 2). CiteAs is similar in spirit to the work on “persuasive technologies” [4], improving research software work by influencing behaviors, but via shaping the incentive environment within which research software is produced, instead of intervening in the development process.

2 CITEAS: A SPECIALIZED SEARCH ENGINE TO CHANGE BEHAVIOR

CiteAs is currently available online via <https://citeas.org/>. Figure 1 demonstrates the front-end interface of CiteAs, which appears as a typical search engine with a search bar. When users enter a search term like a software name, the URL to a “landing page” website, or a unique identifier of the software resource online (e.g., a URL or DOI), CiteAs will recommend a formatted citation according to the search results. The recommended citation can be directly copied or downloaded for use (Figure 3). Users of CiteAs can select from options of citation styles and reference file formats that align with their needs and reference management software.

When generating the citation recommendation, CiteAs also exposes the citation provenance, i.e., the provenance of the citation data it retrieved about the searched-for software. The citation provenance is demonstrated as the step-by-step search path that the CiteAs system has gone through (Figure 4). These steps are built into the search algorithm, covering all kinds of possible locations where different forms of citation requests could be traced. These locations include version-controlled software development platforms such as GitHub; software package repositories such as CRAN and PyPI; or a software project webpage. Examples of software citation requests include a CodeMeta, BibTeX, or CITATION.cff [2] file available online or a language-specific citation request like an R CITATION or DESCRIPTION file.

In the self-explanatory citation provenance, CiteAs demonstrates the locations it reached for the searched-for software and what it has found and not found for recommending citation. For unfound items, CiteAs provides instructions on how to make the item available and discoverable online. Users can visit the source locations to further inspect the citation request or learn from available instructions about how to make a more discoverable citation request.

3 USE CASES

We detail two common use cases of CiteAs to illustrate how it can introduce behavior change to different actors in the scholarly ecosystem. CiteAs can also be useful to stakeholders other than the two major user groups (i.e., researchers and developers of research software) discussed here. We will briefly discuss its other potentials in the final section Future Work.

3.1 Developers get credits for their software work

This case describes how CiteAs enables the effective crediting of research software developers, including researchers who write software themselves, by improving their citation requests. Developers can search for their own software work via CiteAs using search terms they prefer or expect their software users to adopt. When CiteAs returns a citation recommendation, they can explore if it is properly formatted in all available formats for direct use. If the recommended result is not what the developer wants, they can report issues identified via the CiteAs interface. On the other hand, they can examine the citation provenance and self-diagnose if they have made a discoverable citation request for their software work. The citation provenance will expose possible locations online where they can make a citation request retrievable by others. For example,



Figure 3: Sample search results from CiteAs using search term “tidyverse” (name of an R package)—a formatted text citation suggestion. The search result page also provides options for citation downloading formats and citation styles.

if CiteAs discovered a GitHub repository according to the search term used by the developer, but fails to find a citation request of any possible form such as a CodeMeta file, a CITATION/CITATION.cff file, an R DESCRIPTION file, or in the repository README file, etc., CiteAs will indicate that it did not find these possible forms of citation request and give further explanation about these forms. Thus, the developer can learn about how they can make a usable citation request.

It is also possible that the developer prefers that users cite a specific publication relevant to the software other than the software itself. In such cases, the developer can give the details of the publication, or other relevant products (e.g., software documentation; user manual; project website), in their citation request, following the instructions provided by CiteAs.

3.2 Researchers cite the software they used

For researchers, it is critical to provide the details of the software used in research when reporting research activities and outcomes. Citing software in research publications has a positive effect on research transparency and reproducibility. When researchers attempt to cite the software, they need to gather information needed for the citation, including the authors of the software, its unique identifier, version number, access, release date, etc. They then need to format the citation according to the citation style required by their chosen publication venues. Only when the citation is properly included in the body text and bibliography in their paper, can it be indexed by scholarly databases and counted by bibliometric-based research evaluation. However, given the set of choices about how to cite software, the researcher is unlikely to know what type of citation will best recognize the work done to build the software.

By searching for the software they used on CiteAs, researchers can copy the citation recommended by CiteAs based on citation requests made by developers, customized by their selected citation styles. They can also download the recommended citation in formats compatible with reference manager software such as RefWorks, EndNote, and BibTeX, or connect to Zotero for further use in research writings. The citation provenance can point to online access of the software or its relevant work, giving further reference to researchers and facilitating reproducibility.

Citation Provenance (learn more)

- ✓ Looking in the user input, we found a link to a webpage <http://yt-project.org>
- ✗ Looking in the webpage, we didn't find a link to a cite-as relation header
- ✗ Looking in the webpage, we didn't find a DOI. DOI API response
- ✗ Looking in the webpage, we didn't find ArXiv page
- ✓ Looking in the webpage, we found a link to a GitHub repository main page <https://github.com/yt-project/yt>
- ✗ Looking in the GitHub repository main page, we didn't find a DOI. DOI API response
- ✗ Looking in the GitHub repository main page, we didn't find CodeMeta file
- ✓ Looking in the GitHub repository main page, we found a link to a CITATION file <https://raw.githubusercontent.com/yt-project/yt/main/CITATION>
- ✓ Looking in the CITATION file, we found a DOI. DOI API response <https://doi.org/10.1088/0067-0049/192/1/9>
- ✓ Parsing the DOI API response, we found The citation metadata

Figure 4: CiteAs returns citation provenance—its search path is shown, indicating found and missing items. Users can access the citation provenance as well as guidance on missing items so that they can learn to make better citation requests.

It is possible that CiteAs returns an inadequate citation recommendation when citation request is not available online. But it is for this very reason that CiteAs provides a common interface to users and developers of research software, so that their efforts can be coordinated by the same source of citation information.

4 FUTURE WORK

CiteAs is not only able to facilitate coordinated citation efforts by researchers and software developers toward effective crediting of research software work, but also to serve as an information portal and discovery service for research software. One surprise of CiteAs is that users have reported using it to learn about the software they are using, seeking “original academic papers” presenting the design and epistemological intent of the software they are using, and then using forward citation chaining to discover other literature. Thus, it is also broadly useful for other stakeholders of the scholarly ecosystem, including funders, research institutions, scholarly publishers, and other scholarly infrastructures such as software repositories, registries, indexers, and reference managers.

To fulfill its potential, we are adding more functionality to CiteAs. Figure 5 shows the demo of a new feature that informs users how the searched-for software has been mentioned in existing research publications. This feature will be supported by a software knowledge base built on a gold-standard dataset of nearly 5,000 software mentions [3] that recognizes software mentioned in the research literature as well as its authorship, programming language, operating environment, license, etc. This feature can further support research software developers to understand how effective their requests for citation have been and to make an evidenced argument for credit in their local institutional environment.

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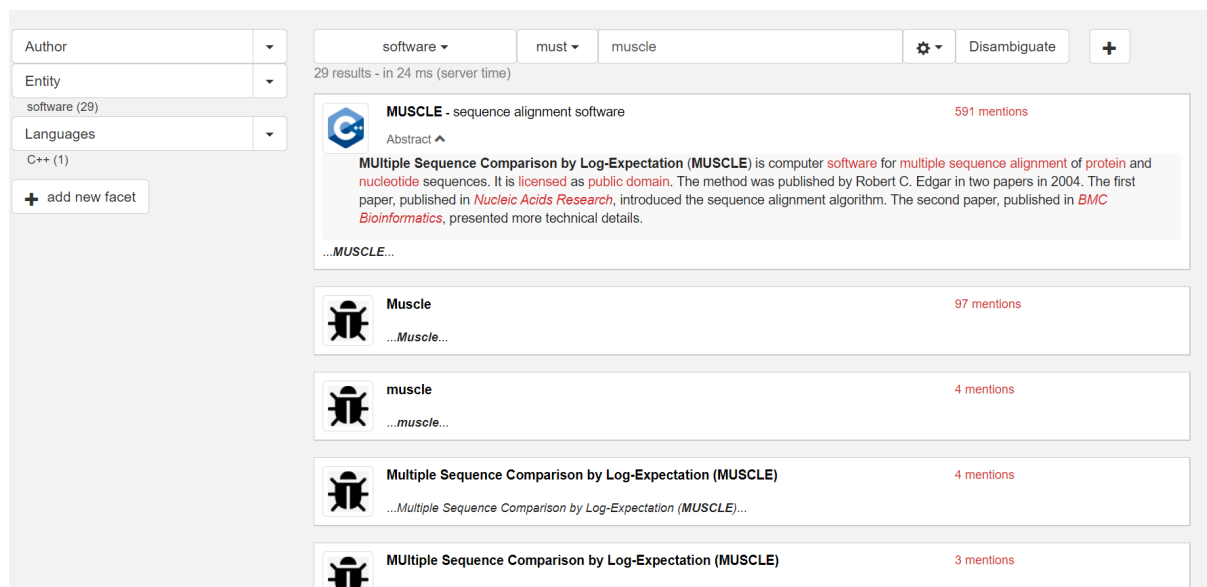


Figure 5: The prototype interface for the software citation recognition functionality—part of the ongoing work of CiteAs